

2NC

We will concede that voting for us means nuclear war per their MacAskill evidence

Nuke war does not equal extinction

[1] - Island populations will repopulate

Turchin-18 (Alexey Turchin, 4/19/2018, Islands as refuges for surviving global catastrophes,

<https://www.emerald.com/insight/content/doi/10.1108/FS-04-2018-0031/full/html?fullSc=1&mbSc=1&fullSc=1//JZ>) In such conditions, proper human perspiration becomes impossible (Sherwood and Huber, 2010), and there will likely be increased mortality and morbidity because of tropical diseases. If temperatures later returned to normal – either naturally or through climate engineering – the rest of the Earth could be repopulated. “Swiss Family Robinsons” survive on a tropical island, unnoticed by a military robot “mutiny”. Most AI researchers ignore medium-term AI risks, which are neither near-term risks, like unemployment, nor remote risks, like AI superintelligence. But a large drone army – if one were produced – could receive a wrong command or be infected by a computer virus, leading it to attack people indiscriminately. Remote islands without robots could provide protection in this case, allowing survival until such a drone army ran out of batteries, fuel, ammunition or other supplies: Primitive tribe survives civilizational collapse. The inhabitants of North Sentinel Island, near the Andaman Islands in the Indian Ocean, are hostile and uncontacted. **The Sentinelese survived the 2004 Indian Ocean tsunami**

apparently unaffected (Voanews, 2009), and if the rest of humanity disappear, they might well continue their existence without change. Tropical Island survives extreme global nuclear winter and glaciation event. Were a nuclear, bolide impactor or volcanic “winter” scenario to unfold, these islands would remain surrounded by Warm Ocean, and local volcanism or other energy sources might provide heat, energy and food. Such island refuges may have helped life on Earth survive during the “Snowball Earth” event in Earth’s distant past (Hoffman et al., 1998). Remote island base for project “Yellow submarine”. Some catastrophic risks such as a gamma ray burst, a global nuclear war with high radiological contamination or multiple pandemics might be best survived underwater in nuclear submarines (Turchin and Green, 2017). However, after a catastrophe, the submarine with survivors would eventually need a place to dock, and an island with some prepared amenities would be a reasonable starting point for rebuilding civilization. Bunker on remote island. For risks which include multiple or complex catastrophes, such as a bolide impact, extreme volcanism, tsunamis, multiple pandemics and **nuclear war with radiological contamination, island refuges could be strengthened with bunkers.** **Richard Branson survived hurricane Irma on his own island in 2017 by seeking refuge in his concrete wine cellar (Clifford, 2017).** **Bunkers on islands would have higher survivability compared to those close to population centers** as they will be neither a military target nor as accessible to looters or unintentionally dangerous (e.g. infected) refugees. These bunkers could potentially be connected to water sources by underwater pipes, and passages could provide cooling, access and even oxygen and food sources.

[2] We've nuked ourselves thousands of times and nothing has happened

Eken-17 (Mattias Eken, 3/14/2017, The understandable fear of nuclear weapons doesn't match reality,

[//JZ\)](https://theconversation.com/the-understandable-fear-of-nuclear-weapons-doesnt-match-reality-73563)

Although the warhead involved in the incident didn't detonate, Schlosser claims that "if it had, much of Arkansas would be gone". But that's not quite the case. The nine-megaton thermonuclear warhead on the Titan II missile had a blast radius of 10km, or an area of about 315km². The state of Arkansas spreads over 133,733km², meaning the weapon would have caused destruction across 0.2% of the state. That would naturally have been a terrible outcome, but certainly not the catastrophe that Schlosser evokes. Claims exaggerating the effects of nuclear weapons have become commonplace, especially after the September 11 terrorist attacks in 2001. In the early War on Terror years, Richard Lugar, a former US senator and chair of the Senate Foreign Relations Committee, argued that terrorists armed with nuclear weapons pose an existential threat to the Western way of life. What he failed to explain is how. **It is by no means certain that a single nuclear**

detonation (or even several) would do away with our current way of life. Indeed, we're still

here despite having nuked our own planet more than 2,000 times – a tally expressed beautifully in this video

by Japanese artist Isao Hashimoto). **While the 1963 Limited Test Ban Treaty forced nuclear tests**

underground, around 500 of all the nuclear weapons detonated were unleashed in the Earth's

atmosphere. This includes the world's largest ever nuclear detonation, the 57-megaton bomb

known as Tsar Bomba, detonated by the Soviet Union on October 30 1961. Tsar Bomba was more than 3,000 times more powerful than the bomb dropped on Hiroshima. That is immense destructive power – but as one physicist explained, it's only "one-thousandth the force of an earthquake, one-thousandth the force of a hurricane". The Damascus incident proved how incredibly hard it is to set off a nuclear bomb and the limited effect that would have come from just one warhead detonating. Despite this, some scientists have controversially argued that an even limited all-out nuclear war might lead to a so-called nuclear winter, since the smoke and debris created by very large bombs could block out the sun's rays for a considerable amount of time. To inflict such ecological societal annihilation with weapons alone, we would have to detonate hundreds if not thousands of thermonuclear devices in a short time. Even in such extreme conditions, the area actually devastated by the bombs would be limited: for example, 2,000 one-megaton explosions with a destructive radius of five miles each would directly destroy less than 5% of the territory of the US. Of course, if the effects of nuclear weapons have been greatly exaggerated, there is a very good reason: since these weapons are indeed extremely dangerous, any posturing and exaggerating which intensifies our fear of them makes us less likely to use them. But it's important, however, to understand why people have come to fear these weapons the way we do. After all, nuclear weapons are here to stay; they can't be "un-invented". If we want to live with them and mitigate the very real risks they pose, we must be honest about what those risks really are. Overegging them to frighten ourselves more than we need to keeps nobody safe.

[3] Tense geopolitics lead politicians and scientists to over exaggerate the impacts of nuclear war. This epistemologically indicts their evidence.

Eken-17 (Mattias Eken, 3/14/2017, The understandable fear of nuclear weapons doesn't match reality,

[//JZ\)](https://theconversation.com/the-understandable-fear-of-nuclear-weapons-doesnt-match-reality-73563) **Nuclear weapons are**

unambiguously the most destructive weapons on the planet. Pound for pound, they are the

most lethal weapons ever created, capable of killing millions. Millions live in fear that these

weapons will be used again, with all the potential consequences. However, the destructive

power of these weapons has been vastly exaggerated, albeit for good reasons. Public fear of nuclear

weapons being used in anger, whether by terrorists or nuclear-armed nations, has [risen once again](#) in recent years. **This is in no**

small part thanks to the current political climate between states such as the US and Russia

and the various nuclear tests conducted by North Korea. But whenever we talk about nuclear

weapons, it's easy to get carried away with doomsday scenarios and apocalyptic language. As the historian Spencer Weart once argued: "You say 'nuclear bomb' and everybody immediately thinks of the end of the world." Yet the means necessary to produce a nuclear bomb, let alone set one off, remain incredibly complex – and while the damage that would be done if someone did in fact detonate one might be very serious indeed, the chances that it would mean "the end of the world" are vanishingly small. In his 2013 book *Command and Control*, the author Eric Schlosser tried to scare us into perpetual fear of nuclear weapons by recounting stories of near misses and accidents involving nuclear weapons. One such event, the 1980 Damascus incident, saw a Titan II intercontinental ballistic missile explode at its remote Arkansas launch facility after a maintenance crew accidentally ruptured its fuel tank. Although the warhead involved in the incident didn't detonate, Schlosser claims that "if it had, much of Arkansas would be gone".

[4] Food storages solve

Ladish-20 (Jeffrey Ladish, 11/06/2020, Nuclear war is unlikely to cause human extinction, LessWrong, <https://www.lesswrong.com/posts/sT6NxXso6Z9xjS7o/nuclear-war-is-unlikely-to-cause-human-extinction//IJZ>) Two different analyses are required to calculate the chances of human extinction from nuclear winter. The first is the analysis of the climate change that could result from a nuclear war, and the second is the adaptive capacity of human groups to these climate changes. I have not seen an in depth analysis of the latter, but I believe such an assessment would be worthwhile. My own guess is that humans are capable of surviving far more severe climate shifts than those projected in nuclear winter scenarios. Humans are more robust than most any other mammal to drastic changes in temperature, as evidenced by our global range, even in pre-historic times. While a loss of most agriculture would likely kill most people on earth, modern technology would enable some populations to survive. Great stores of food currently exist in the world, and it is likely that some of these would be seized and protected by small groups, providing enough food to last for years. While even such populations with food stores wouldn't have enough to survive for 10-15 years, such food stores would give groups time to adapt to new food sources. The organization ALLFED has explored a number of alternative food sources that could keep populations alive in the event of a nuclear war or other large solar disruption, and I expect great necessity to drive the discovery of even more in the event of such a disaster. B: The Robock group's models are probably overestimating the risk The nuclear winter model at its simplest: Nuclear detonations → Fires in cities → Firestorms in cities → Lofted black carbon into the upper atmosphere → black carbon persists in upper atmosphere, reflecting sunlight and causes massive cooling Each step is required in order for the effect to occur. If nuclear war causes massive fires in cities but does not lead to firestorms that loft particles, then no long term cooling is going to occur.

[5] Farming without sunlight solves

Swenson-22 (Shea Swenson, 7/28/2022, Cultivating Crops, No Sun Required, Modern Farmer, <https://modernfarmer.com/2022/07/artificial-photosynthesis//IJZ>) Photosynthesis is one of life's most quintessential recipes. Plants combine sunlight, water and carbon dioxide to produce oxygen, along with nutrients the organisms need to grow. Now, scientists have found a way to remove sunlight out of the equation. In a new study conducted by the University of California, Riverside and the University of Delaware, researchers have successfully grown plants using artificial photosynthesis—meaning the plants have no need for contact with sunlight to grow effectively. Instead, the plants rely on acetate—a chemical compound that plants can use to grow—produced in a sunless lab. Researchers got around the need for

sunlight by using an electrocatalytic process in which water, electricity and carbon dioxide are used to produce acetate. The acetate then replaces the plant's need for glucose, which plants create and consume as part of traditional photosynthesis. Once the acetate is made, researchers add it to vials, in which the plants grow hydroponically. For the process, researchers use a tool called a carbon dioxide electrolyzer. According to a press release, electrolyzers are "devices that use electricity to convert raw materials like carbon dioxide into useful molecules and products." In this case, the electrolyzer uses electricity to transform carbon dioxide into acetate. The technology isn't entirely operated without sun. In this case, the electricity captured to create the acetate comes from solar panels. **The plants, however, still don't soak in any of the sun's rays. With acetate as a substitute food source, the plants were not only able to grow in complete darkness, some even thrived. According to the study, for some of the plants, the in-the-dark process was up to 18 times more effective than photosynthesis with sunlight. Unlike the artificial process, biological photosynthesis isn't exactly energy efficient. According to the study, plants typically end up with just one percent of the energy found in sunlight.** "This technology is a more efficient method of turning solar energy into food, as compared to food production that relies on biological photosynthesis," said Elizabeth Hann, a doctoral candidate in the Jinkerson Lab and co-lead author of the study, in a press release. The researchers had success with food-bearing crops such as rice, jalapeños, tomatoes, green peas and lettuce, in addition to growing yeast, algae and mushroom-producing fungus. The innovation leaves the door open to more successful indoor growing, with less need for large outdoor land use when it comes to crop growth in the future. **In addition to more effective indoor growing here on Earth, the process provides a possible solution for out-of-this-world cultivation.** Robert Jinkerson, UC Riverside assistant professor of chemical and environmental engineering and corresponding author of the study, said: **"Using artificial photosynthesis approaches to produce food could be a paradigm shift for how we feed people. By increasing the efficiency of food production, less land is needed, lessening the impact agriculture has on the environment. And for agriculture in non-traditional environments, like outer space, the increased energy efficiency could help feed more crew members with less inputs."**

No Rebuilding

[1] Industrial civilization will never recover from nuclear war

Dartnell-15 (Lewis Dartnell, 4/13/2015, Could we reboot a modern civilisation without fossil fuels? – Lewis Dartnell, <https://aeon.co/essays/could-we-reboot-a-modern-civilisation-without-fossil-fuels//12>)

Imagine that the world as we know it ends tomorrow. There's a global catastrophe: a pandemic virus, an asteroid strike, or perhaps a nuclear holocaust. The vast majority of the human race perishes. Our civilisation collapses. The post-apocalyptic survivors find themselves in a devastated world of decaying, deserted cities and roving gangs of bandits looting and taking by force. Bad as things sound, that's not the end for humanity. We bounce back. Sooner or later, peace and order emerge again, just as they have time and again through history. Stable communities take shape. They begin the agonising process of rebuilding their technological base from scratch. But here's the question: how far could such a society rebuild? Is there any chance, for instance, that a post-apocalyptic society could reboot a technological civilisation? Let's make the basis of this thought experiment a little more specific. Today, **we have already consumed the most easily**

drainable crude oil and, particularly in Britain, much of the shallowest, most readily mined deposits of coal. Fossil fuels are central to the organisation of modern industrial society, just as they were central to its development. Those, by the way, are distinct roles: even if we could somehow do without fossil fuels now (which we can't, quite), it's a different question whether we could have got to where we are without ever having had them. So, would a society starting over on a planet stripped of its fossil fuel deposits have the chance to progress through its own Industrial Revolution? Or to phrase it another way, what might have happened if, for whatever reason, the Earth had never acquired its extensive underground deposits of coal and oil in the first place? Would our progress necessarily have halted in the 18th century, in a pre-industrial state? It's easy to underestimate our current dependence on fossil fuels. In everyday life, their most visible use is the petrol or diesel pumped into the vehicles that fill our roads, and the coal and natural gas which fire the power stations that electrify our modern lives. But we also rely on a range of different industrial materials, and in most cases, high temperatures are required to transform the stuff we dig out of the ground or harvest from the landscape into something useful. You can't smelt metal, make glass, roast the ingredients of concrete, or synthesise artificial fertiliser without a lot of heat. It is fossil fuels – coal, gas and oil – that provide most of this thermal energy. In fact, the problem is even

worse than that. Many of the chemicals required in bulk to run the modern world, from pesticides to plastics, derive from the diverse organic compounds in crude oil. Given the dwindling reserves of crude oil left in the world, it could be argued that the most wasteful use for this limited resource is to simply burn it. We should be carefully preserving what's left for the vital repertoire of valuable organic compounds it offers. But my topic here is not what we should do now. Presumably everybody knows that we must transition to a low-carbon economy one way or another. No, I want to answer a question whose interest is (let's hope) more theoretical. Is the emergence of a technologically advanced civilisation necessarily contingent on the easy availability of ancient energy? Is it possible to build an industrialised civilisation without fossil fuels? And the answer to that question is: maybe – but it would be extremely difficult. Let's see how. We'll start with a natural thought: Many of our alternative energy technologies are already highly developed. Solar panels, for example, represent a good option today, and are appearing more and more on the roofs of houses and businesses. It's tempting to think that a rebooted society could simply pick up where we leave off. Why couldn't our civilisation 2.0 just start with renewables? Well, it could, in a very limited way if you find yourself among the survivors in a post-apocalyptic world, you could scavenge enough working solar panels to keep your lifestyle electrified for a good long while. Without moving parts, photovoltaic cells require little maintenance and are remarkably resilient. They do deteriorate over time, though, from moisture penetrating the casing and from sunlight itself degrading the high-purity silicon layers. The electricity generated by a solar panel declines by about 1 per cent every year so, after a few generations, all our hand-me-down solar panels will have degraded to the point of uselessness. Then what? New ones would be fiendishly difficult to create from scratch. Solar panels are made from thin slices of extremely pure silicon, and although the raw material is common sand, it must be processed and refined using complex and precise techniques – the same technological capabilities, more or less, that we need for modern semiconductor electronics components. These techniques took a long time to develop, and would presumably take a long time to recover. So photovoltaic solar power would not be within the capability of a society early in the industrialisation process. Perhaps, though, we were on the right track by starting with electrical power. Most of our renewable-energy technologies produce electricity. In our own historical development, it so happens that the core phenomena of electricity were discovered in the first half of the 1800s, well after the early development of steam engines. Heavy industry was already committed to combustion-based machinery, and electricity has largely assumed a subsidiary role in the organisation of our economies ever since. But could that sequence have run the other way? Is there some developmental requirement that thermal energy must come first? On the face of it, it's not beyond the bounds of possibility that a progressing society could construct electrical generators and couple them to simple windmills and waterwheels, later progressing to wind turbines and hydroelectric dams. In a world without fossil fuels, one might envisage an electrified civilisation that largely bypasses combustion engines, building its transport infrastructure around electric trains and trams for long-distance and urban transport. I say 'largely'. We couldn't get round it all together. When it comes to generating the white heat demanded by modern industry, there are few good options but to burn stuff. While the electric motor could perhaps replace the coal-burning steam engine for mechanical applications, society, as we've already seen, also relies upon thermal energy to drive the essential chemical and physical transformations it needs. How could an industrialising society produce crucial building materials such as iron and steel, brick, mortar, cement and glass without resorting to deposits of coal? You can of course create heat from electricity. We already use electric ovens and kilns. Modern arc furnaces are used for producing cast iron or recycling steel. The problem isn't so much that electricity can't be used to heat things, but that for meaningful industrial activity you've got to generate prodigious amounts of it, which is challenging using only renewable energy sources such as wind and water. An alternative is to generate high temperatures using solar power directly. Rather than relying on photovoltaic panels, concentrated solar thermal farms use giant mirrors to focus the sun's rays onto a small spot. The heat concentrated in this way can be exploited to drive certain chemical or industrial processes, or else to raise steam and drive a generator. Even so, it is difficult (for example) to produce the very high temperatures inside an iron-smelting blast furnace using such a system. What's more, it goes without saying that the effectiveness of concentrated solar power depends strongly on the local climate. No, when it comes to generating the white heat demanded by modern industry, there are few good options but to burn stuff. But that doesn't mean the stuff we burn necessarily has to be fossil fuels. Let's take a quick detour into the pre-history of modern industry. Long before the adoption of coal, charcoal was widely used for smelting metals. In many respects it is superior: charcoal burns hotter than coal and contains far fewer impurities. In fact, coal's impurities were a major delaying factor on the Industrial Revolution. Released during combustion, they can taint the product being heated. During smelting, sulphur contaminants can soak into the molten iron, making the metal brittle and unsafe to use. It took a long time to work out how to treat coal to make it useful for many industrial applications. And, in the meantime, charcoal worked perfectly well. And then, well, we stopped using it. In retrospect, that's a pity. When it comes from a sustainable source, charcoal burning is essentially carbon-neutral, because it doesn't release any new carbon into the atmosphere – not that this would have been a consideration for the early industrialists. But charcoal-based industry didn't die out altogether. In fact, it survived to flourish in Brazil. Because it has substantial iron deposits but few coalmains, Brazil is the largest charcoal producer in the world and the ninth biggest steel producer. We aren't talking about a cottage industry here, and this makes Brazil a very encouraging example for our thought experiment. The trees used in Brazil's charcoal industry are mainly fast-growing eucalyptus, cultivated specifically for the purpose. The traditional method for creating charcoal is to pile chopped staves of air-dried timber into a great dome-shaped mound and then cover it with turf or soil to restrict airflow as the wood smoulders. The Brazilian enterprise has scaled up this traditional craft to an industrial operation. Dried timber is stacked into neat, cylindrical kilns, built of brick or masonry and arranged in long lines so that they can be easily filled and unloaded in sequence. The largest sites can sport hundreds of such kilns. Once filled, their entrances are sealed and a fire is lit from the top. The skill in charcoal production is to allow just enough air into the interior of the kiln. There must be enough combustion heat to drive out moisture and volatiles and to pyrolyse the wood, but not so much that you are left with nothing but a pile of ash. The kiln attendant monitors the state of the burn by carefully watching the smoke seeping out of the top, opening air holes or sealing with clay as necessary to regulate the process. Brazil shows how the raw materials of modern civilisation can be supplied without reliance on fossil fuels. Good things come to those who wait, and this wood pyrolysis process can take up to a week of carefully controlled smouldering. The same basic method has been used for millennia. However, the ends to which the fuel is put are distinctly modern. Brazilian charcoal is trucked out of the forests to the country's blast furnaces where it is used to transform ore into pig iron. This pig iron is the basic ingredient of modern mass-produced steel. The Brazilian product is exported to countries such as China and the US where it becomes cars and trucks, sinks, bathtubs, and kitchen appliances. Around two-thirds of Brazilian charcoal comes from sustainable plantations, and so this modern-day practice has been dubbed 'green steel'. Sadly, the final third is supplied by the non-sustainable felling of primary forest. Even so, the Brazilian case does provide an example of how the raw materials of modern civilisation can be supplied without reliance on fossil fuels. Another, related option might be wood gasification. The use of wood to provide heat is as old as mankind, and yet simply burning timber only uses about a third of its energy. The rest is lost when gases and vapours released by the burning process blow away in the wind. Under the right conditions, even smoke is combustible. We don't want to waste it. Better than simple burning, then, is to drive the thermal breakdown of the wood and collect the gases. You can see the basic principle at work for yourself just by lighting a match. The luminous flame isn't actually touching the matchwood: it dances above, with a clear gap in between. The flame actually feeds the hot gases given off as the wood breaks down in the heat, and the gases combust only once they mix with oxygen from the air. Matches are fascinating when you look them closely. Wartime gasifier cars could achieve about 1.5 miles per kilogram. Today's designs improve upon this. To release these gases in a controlled way, bake some timber in a closed container. Oxygen is restricted so that the wood doesn't simply catch fire. Its complex molecules decompose through a process known as pyrolysis, and then the hot carbonised lumps of charcoal at the bottom of the container react with the breakdown products to produce flammable gases such as hydrogen and carbon monoxide. The resultant 'producer gas' is a versatile fuel: it can be stored or piped for use in heating or street lights, and is also suitable for use in complex machinery such as the internal combustion engine. More than a million gasifier-powered cars across the world kept civilian transport running during the oil shortages of the Second World War. In occupied Denmark, 95 per cent of all tractors, trucks and fishing boats were powered by wood-gas generators. The energy content of about 3 kg of wood (depending on its dryness and density) is equivalent to a litre of petrol, and the fuel consumption of a gasifier-powered car is given in miles per kilogram of wood rather than miles per gallon. Wartime gasifier cars could achieve about 1.5 miles per kilogram. Today's designs improve upon this. But you can do a lot more with wood gases than just keep your vehicle on the road. It turns out to be suitable for any of the manufacturing processes needing heat that we looked at before, such as kilns for lime, cement or bricks. Wood gas generator units could easily power agricultural or industrial equipment, or pumps. Sweden and Denmark are world leaders in their use of sustainable forests and agricultural waste for turning the steam turbines in power stations. And once the steam has been used in their 'Combined Heat and Power' (CHP) electricity plants, it is piped to the surrounding towns and industries to heat them, allowing such CHP stations to approach 90 per cent energy efficiency. Such plants suggest a

marvellous vision of industry wholly weaned from its dependency on fossil fuel. Is that our solution, then? Could our rebooting society run on wood, supplemented with electricity from renewable sources? Maybe so, if the population was

fairly small. But here's the catch. These options all presuppose that our survivors are able to construct efficient steam turbines, CHP stations and internal combustion engines. We know how to do all that, of course – but in the event of a civilisational collapse, who is to say that the knowledge won't be lost? And if it is, what are the chances that our descendants could reconstruct it? In our own history, the first successful application of steam engines was in pumping out coal mines. This was a setting in which fuel was already abundant, so it didn't matter that the first, primitive designs were terribly inefficient. The increased output of coal from the mines was used to first smelt and then forge more iron. Iron components were used to construct further steam engines, which were in turn used to pump mines or drive the blast furnaces at iron foundries. And of course, steam engines were themselves employed at machine shops to construct yet more steam engines. It was only once steam engines were being built and operated that subsequent engineers were able to devise ways to increase their efficiency and shrink fuel demands.

Once the industry is gone we can't get it back

Jacobi-17 (John Jacobi, 5/27/2017, Industrial Civilization Could Not Be Rebuilt, The Wild Will Project.

<https://www.wildwill.net/blog/2017/05/27/industrial-civilization-not-rebuilt//J2>)

A suggestion, for the sake of thought: If industrial civilization collapsed, it probably could not be rebuilt. Civilization would exist again, of course, but industry appears to be a one-time experiment. The astronomer Fred Hoyle, exaggerating

slightly, writes: It has often been said that, if the human species fails to make a go of it here on Earth, some other species will take over the running. In the sense of developing high intelligence this is not correct. We have, or soon will have, exhausted the necessary physical prerequisites so far as this planet is concerned. **With coal gone, oil gone, high-grade metallic ores gone, no species however competent can make the long climb from primitive conditions to high-level technology.** This is a one-shot affair. If we fail, this planetary system fails so far as intelligence is concerned. The same will be true of other planetary systems. On each of them there will be one chance, and one chance only. Hoyle overstates all the limits we actually have to worry about, but there are enough to affirm his belief that **industry is a "one-shot affair."** In other words, if industry collapsed then no matter how quickly scientific knowledge allows societies to progress, technical development will hit a wall because the builders will not have the needed materials. For example, much of the world's land is not arable, and some of the land in use today is only productive because of industrial techniques developed during the agricultural revolution in the 1600s, techniques heavily dependent on oil. Without the systems that sustain industrial agriculture much current farm land could not be farmed; agricultural civilizations cannot exist there, at least until the soil replenishes, if it replenishes. And some resources required for industrial progress, like coal, simply are not feasibly accessible anymore. Tainter writes: . . . major jumps in population, at around A.D. 1300, 1600, and in the late eighteenth century, each led to intensification in agriculture and industry. As the land in the late Middle Ages was increasingly deforested to provide fuel and agricultural space for a growing population, basic heating, cooking, and manufacturing needs could no longer be met by burning wood. A shift to reliance on coal began, gradually and with apparent reluctance. Coal was definitely a fuel source of secondary desirability, being more costly to obtain and distribute than wood, as well as being dirty and polluting. Coal was more restricted in its spatial distribution than wood, so that a whole new, costly distribution system had to be developed. Mining of coal from the ground was more costly than obtaining a quantity of wood equivalent in heating value, and became even more costly as the 54 most accessible reserves of this fuel were depleted. Mines had to be sunk ever deeper, until groundwater flooding became a serious problem. Today, most easily **accessible natural coal reserves are completely depleted.** **Thus, societies in the wake of our imagined collapse would not be able to develop fast enough to reach the underground coal. As a result of these limits, rebuilding industry would take at least thousands of years** — it took 10,000 years the first time around. By the time a civilization reached the point where it could do something about industrial scientific knowledge it probably would not have the knowledge anymore. It would have to develop its sciences and technologies on its own, resulting in patterns of development that would probably look similar to historical patterns. Technology today depends on levels of complexity that must proceed in chronological stages. Solar panels, for example, rely on transportation infrastructure, mining, and a regulated division of labor. And historically **the process of developing into a global civilization includes numerous instances of technical regression.** The natives of Tasmania, for example, went from a maritime society to one that didn't fish, build boats, or make bows and arrows. Rebuilding civilization would also be a bad idea. Most, who are exploited by rather than benefit from industry, would probably not view a rebuilding project as desirable. Even today, though citizens of first-world nations live physically comfortable lives, their lives are sustained by the worse off lives of the rest of the world. "Civilization . . . has operated two ways," Paine writes, "to make one part of society more affluent, and the other more wretched, than would have been the lot of either in a natural state."

[2] Nuclear war prevents tech research

Baum-18 (Seth Baum, 4/19/2018, A Model for the Impacts of Nuclear War, Global Catastrophic Risk Institute SSRN Electronic Journal. Crossref, https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3155983 //JZ) The total impact of nuclear war is a major factor in many important policy questions, but it has gotten little scholarly attention. This paper presents a model for calculating the total impacts of nuclear war. The model includes physical, infrastructural, and social impacts as they affect human lives. The model has five main branches corresponding to the five main types of effects of nuclear weapon detonations: thermal radiation, blast, ionizing radiation, electromagnetic pulse, and human perceptions. Model branches contain extensive detail on each of these effects, including interconnections between them and connections to other major risks including global warming and pandemics. The paper also includes background information on impacts analysis and modeling to help readers understand how to think about the impacts of nuclear war, including discussion of important attributes of nuclear war such as the number and yield of weapons detonated and the location of their detonation. **Another link between nuclear war and other major catastrophes comes from the potential for general malfunction of society shifting work on risky technologies such as artificial intelligence, molecular nanotechnology, and biotechnology. The simplest effect would be for the general malfunction of society to halt work on**

these technologies. In most cases, this would reduce the risk of harm caused by those technologies. As geopolitical tensions rise in nuclear-armed states, scientists are modelling the global impact of nuclear war. The model has five main branches corresponding to the five main types of effects of nuclear weapon detonations: thermal radiation, blast, ionizing radiation, electromagnetic pulse, and human perceptions. Model branches contain extensive detail on each of these effects, including interconnections between them and connections to other major risks including global warming and pandemics. The paper also includes background information on impacts analysis and modeling to help readers understand how to think about the impacts of nuclear war, including discussion of important attributes of nuclear war such as the number and yield of weapons detonated and the location of their detonation.

[3] All experts that could guide reconstruction would be killed.

Wolfson '22 [Richard and Ferenc Dalnoki-Veress; March 2; Benjamin F. Wissler Professor of Physics at Middlebury College, Ph.D. in Physics from Dartmouth College, M.A. in Environmental Studies from the University of Michigan; Scientist-in-Residence at the Center for Nonproliferation Studies of the Middlebury Institute of International Studies, Ph.D. and M.Sc. in High-Energy Physics from Carleton University; The MIT Press Reader, “The Devastating Effects of Nuclear Weapons,”

<https://thereader.mitpress.mit.edu/devastating-effects-of-nuclear-weapons-war/>] pbk Individuals might survive for a while, but what about longer term, and what about society as a whole? Extreme and cooperative efforts would be needed for long-term survival, but would the shocked and weakened survivors be up to those efforts? How would individuals react to watching their loved ones die of radiation sickness or untreated injuries? Would an “everyone for themselves” attitude prevail, preventing the cooperation necessary to rebuild society? How would residents of undamaged rural areas react to the streams of urban refugees flooding their communities? What governmental structures could function in the postwar climate? How could people know what was happening throughout the country? Would international organizations be able to cope?

Staying indoors for 48 hours after a nuclear blast is now recommended. That time allows fallout levels to decay by a factor of 100.

Some students of nuclear war see postwar society in a race against time. An all-out war would have destroyed much of the nation's productive capacity and would have killed many of the experts who could help guide social and physical reconstruction. The war also would have destroyed stocks of food and other materials needed for survival.

On the other hand, the remaining supplies would have to support only the much smaller postwar population. The challenge to the survivors would be to establish production of food and other necessities before the supplies left from before the war were exhausted. Could the war-shocked survivors, their social and governmental structure shattered, meet that challenge? That is a very big nuclear question — so big that it's best left unanswered, since only an all-out nuclear war could decide it definitively.

Offense

The military is developing isomer bombs---testing will destroy the Universe

Bekkum 4 – Gary S. Bekkum, Founder of Spacetime Threat Assessment Report Research, Founder of STARstream Research, Futurist, "American Military is Pursuing New Types of Exotic Weapons", Pravda, 8-30,
http://www.starstreamresearch.com/dark_matters.htm

Recently the British science news journal "New Scientist" revealed that the American military is pursuing new types of exotic bombs - including a new class of isomeric gamma ray weapons. Unlike conventional atomic and hydrogen bombs, the new weapons would trigger the release of energy by absorbing radiation, and respond by re-emitting a far more powerful radiation. In this new category of gamma-ray weapons, a nuclear isomer absorbs x-rays and re-emits higher frequency gamma rays. The emitted gamma radiation has been reported to release 60 times the energy of the x-rays that trigger the effect. The discovery of this isomer triggering is fairly recent, and was first reported in a 1999 paper by an international group of scientists. Although this controversial development has remained fairly obscure, it has not been hidden from the public. Beyond the visible part of defense research is an immense underground of secret projects considered so sensitive that their very existence is denied. These so-called "black budget programs" are deliberately kept from the public eye and from most political leaders. CNN recently reported that in the United States the black budget projects for 2004 are being funded at a level of more than 20 billion dollars per year. In the summer of 2000 I contacted Nick Cook, the former aviation editor and aerospace consultant to Jane's Defence Weekly, the international military affairs journal. Cook had been investigating black budget super-secret research into exotic physics for advanced propulsion technologies. I had been monitoring electronic discussions between various American and Russian scientists theorizing about rectifying the quantum vacuum for advanced space drive. Several groups of scientists, partitioned into various research organizations, were exploring what NASA calls "Breakthrough Propulsion Physics" - exotic technologies for advanced space travel to traverse the vast distances between stars. Partly inspired by the pulp science fiction stories of their youth, and partly by recent reports of multiple radar tracking tapes of unidentified objects performing impossible maneuvers in the sky, these scientists were on a quest to uncover the most likely new physics for star travel. The NASA program was run by Marc Millis, financed under the Advanced Space Transportation Program Office (ASTP). Joe Firmage, then the 28-year-old Silicon Valley CEO of the three billion dollar Internet firm US Web, began to fund research in parallel with NASA. Firmage hired a NASA Ames nano-technology scientist, Creon Levit, to run the "International Space Sciences Organization", a move which apparently alarmed the management at NASA. The San Francisco based Hearst Examiner reported that NASA's Office of Inspector General assigned Special Agent Keith Tate to investigate whether any proprietary NASA technology might have been leaking into the private sector. Cook was intrigued when I pointed out the apparent connections between various private investors, defense contractors, NASA, INSCOM (American military intelligence), and the CIA. While researching exotic propulsion technologies Cook had heard rumors of a new kind of weapon, a "sub-quantum atomic bomb", being whispered about in what he called the "dark halls" of defense research. Sub-quantum physics is a controversial re-interpretation of quantum theory, based on so-called pilot wave theories, where an information field controls quantum particles. The late Professor David Bohm showed that the predictions of ordinary quantum mechanics could be recast into a pilot wave information theory. Recently Anthony Valentini of the Perimeter Institute has suggested that ordinary quantum theory may be a special case of pilot wave theories, leaving open the possibility of new and exotic non-quantum technologies. Some French, Serbian and Ukrainian physicists have been working on new theories of extended electrons and solitons, so perhaps a sub-quantum bomb is not

entirely out of the question. Even if the rumors of a sub-quantum bomb are pure fantasy, there is no question that mainstream physicists seriously contemplate a phase transition in the quantum vacuum as a real possibility. The quantum vacuum defies common sense, because empty space in quantum field theory is actually filled with virtual particles. These virtual particles appear and disappear far too quickly to be detected directly, but their existence has been confirmed by experiments that demonstrate their influence on ordinary matter. "Such research should be forbidden!" In the early 1970's Soviet physicists were concerned that the vacuum of our universe was only one possible state of empty space. The fundamental state of empty space is called the "true vacuum". Our universe was thought to reside in a "false vacuum", protected from the true vacuum by "the wall of our world". A change from one vacuum state to another is known as a phase transition. This is analogous to the transition between frozen and liquid water. Lev Okun, a Russian physicist and historian recalls Andrei Sakharov, the father of the Soviet hydrogen bomb, expressing his concern about research into the phase transitions of the vacuum. If the wall between vacuum states was to be breached, calculations showed that an unstoppable expanding bubble would continue to grow until it destroyed our entire universe! Sakharov declared that "Such research should be forbidden!" According to Okun, Sakharov feared that an experiment might accidentally trigger a vacuum phase transition.

Scientists are closing in on Absolute Zero

Lea '22 [Robert, science journalist in the U.K. whose articles have been published in Physics World, New Scientist, Astronomy Magazine, All About Space, Newsweek and ZME Science. Sept 8, 2022, Space, "Scientists create coldest matter in the universe in a lab", <https://www.space.com/coldest-matter-in-universe-created-in-lab>] acc 7/23/2024 [tvisha]

A team of researchers has cooled matter to within a billionth of a degree of absolute zero, colder than even the deepest depths of space, far away from any stars. Interstellar space never gets this cold due to the fact that it is evenly filled with the cosmic microwave background (CMB), a form of radiation left over from an event that occurred shortly after the Big Bang when the universe was in its infancy. The chilled matter is even colder than the coldest known region of space, the Boomerang Nebula, located 3,000 light-years from Earth, which has a temperature of just one degree above absolute zero. The experiment, run at the University of Kyoto in Japan and used fermions, which is what particle physicists call any particle that makes up matter, including electrons, protons and neutrons. The team cooled their fermions – atoms of the element ytterbium – to around a billionth of a degree above absolute zero, the hypothetical temperature at which all atomic movement would cease. Sponsored Links Play this game for 1 minute and see why everyone is addicted. Raid Shadow Legends "Unless an alien civilization is doing experiments like these right now,

anytime this experiment is running at Kyoto University it is making the coldest fermions in the universe," Rice University researcher Kaden Hazzard, who took part in the study, said in a statement. Related: Did this newfound particle form the universe's dark matter? Click here for more Space.com videos... CLOSE The team used lasers to cool the matter by restricting the motion of 300,000 atoms within an optical lattice. The experiment simulates a model of quantum physics first proposed in 1963 by theoretical physicist, John Hubbard. The so-called Hubbard model allows atoms to demonstrate unusual quantum properties including collective behavior between electrons like superconduction (the ability to conduct electricity without energy loss). "The payoff of getting this cold is that the physics really changes," Hazzard said. "The physics starts to become more quantum mechanical, and it lets you see new phenomena." The 'fossil' radiation that keeps space warm Interstellar space can never get this cold because of the presence of the CMB. This evenly spread and uniform radiation was created by an event during the initial rapid expansion of the universe shortly after the Big Bang, the so-called last scattering. During the last scattering, electrons started to bond with protons, forming the first atoms of the lightest existing element hydrogen. As a result of this atom formation, the universe rapidly lost its loose electrons. And because electrons scatter photons, the universe had been opaque to light before the last scattering. With electrons bound up with protons in these first hydrogen atoms, photons could suddenly travel freely, making the universe transparent to light. The last scattering also marked the last moment at which fermions like protons and photons had the same temperature. As a result of the last scattering, photons filled the universe at a specific temperature of 2.73 Kelvin, which equals minus 454.76 degrees Fahrenheit (minus 270.42 degrees Celsius) which is just 2.73 degrees above absolute zero — 0 Kelvin or minus 459.67 degrees F (minus 273.15 degrees C). Boomerang nebula - coldest place in the universe The Boomerang nebula is the coldest place in the universe (Image credit: ESA/NASA) RELATED STORIES: — The history of the universe: Big Bang to now in 10 easy steps — What happened before the Big Bang? — Dark matter from 12 billion years ago detected for the 1st time There is one region in the known universe, the Boomerang Nebula, a cloud of gas that surrounds a dying star in the constellation of Centaurus, which is even colder than the rest of the universe — around 1 Kelvin or minus 457.6 °F (minus 272° C). Astronomers believe the Boomerang Nebula is being cooled by cold, expanding gas spat out by the dying star at the nebula's center. But even the Boomerang Nebula can't compete with the temperatures of the ytterbium atom in the latest experiment. The team behind this experiment is currently working on developing the first tools capable of measuring the behavior that arises a billionth of a degree above absolute zero. "These systems are pretty exotic and special, but the hope is that by studying and understanding them, we can identify the key ingredients that need to be there in real materials," Hazzard concluded.

That destroys the Universe

Hohmann 16 [Creator and Editor of Moonwards. May 29, 2016, World Building, "Temperatures Below Zero Degrees Kelvin,"

<https://worldbuilding.stackexchange.com/questions/42510/temperatures-below-zero-degrees-kelvin>] acc 7/24/2024 [tvisha]

Temperature is a measure of how much the atoms vibrate and move (e.g. kinetic energy). 0 Kelvin is defined as the point where the atoms do not vibrate any more at all, so you can not really have sub-kelvin temperatures, as it is not possible to move less than standing totally still. If you somehow discover that atoms at what we now believe is absolute zero actually move a little, your new temperature is just defined as the new 0 Kelvin. A second consideration is that negative degrees Kelvin allows for heat flow that reduces entropy, a pretty important violation of physics, as that allows for infinite energy, perpetual motion machines, etc. In short, negative absolute temperatures can not exist in reality, and in a sci-fi setting, it is likely to have a lot of unexpected consequences, having a good chance to somehow

destroy the universe

Time Travel

Elmi 18 – Awes Faghi Elmi, Contributing Writer at n'world Publications, BS in Forensic Science from London South Bank University, Extended Diploma in Physics with Distinction from Leyton Sixth Form College, Futurist, "Technological Progress Might Make Possible Time Travel And Teleportation", Medium, 8-13,
<https://medium.com/nworld-publications/technological-progress-might-make-possible-time-travel-and-teleportation-45176c3c89bc> [typo edited]

Can we travel through time? This is a question that many people ask their-selves. This question has occurred many times. It is said that **time travel is possible** and in fact it is. The key things needed to travel through time are speed and kinetic energy. Einstein's theory also known as the theory of relativity can be used [to] to understand how to deal with travelling to the future. Einstein showed that travelling forward in time is easy. According to Einstein's theory of relativity, time passes at different rates for people who are moving relative to one another although the effect only becomes large when you get close to the speed of light. Time travel sometime can cause side effects called paradoxes. These paradoxes can occur especially when going back in time. As if only one thing even the minimum of the details can change something big may happen in the future. Another scientist who believes that time travel is possible after Einstein is Brian Cox who as Einstein believes that we are only going to be able to travel in the future. This obviously would happen if having a super-fast machine that allows you to go into the future. Cox also agrees on Einstein's theory of relativity which states that to travel forward in time, something needs to reach speeds close to the speed of light. As it approaches these speeds, time slows down but only for that specific object. **They both think as said, that time travel to the future is possible however travelling back in time is impossible, as something must be really as fast as the speed of light. This however** for some scientists **can be wrong. They state that with the technology that we have now it could be possible to build some sort of machine who will actually be able to travel in both future and past**. A wormhole as shown in the image is a theoretical passage through space-time that could create shortcuts for long journeys across the universe. Wormholes are predicted by the theory of general relativity. However, wormholes bring with them the dangers of sudden collapse, high radiation and dangerous contact with exotic matter. The public knows that time travel is possible but humans at the moment are not able to. However other sources except theories of the past are currently trying to develop a way of time travel. The audience actually cannot wait that this will happen as many media state, such as BBC. Many TV programmes talk about both time travel and teleportation. Teleportation Quantum teleportation is a process by which quantum information (e.g. the exact state of an atom or photon) can be transmitted (exactly, in principle) from one location to another, with the help of classical communication and previously shared quantum entanglement between the sending and receiving location. Albert Einstein, one of the best intelligent minds in the world, believed that teleportation using quantum mechanics was and will be impossible. This however had a breakthrough few scientists in Delft University were finally able to learn how to use the quantum mechanics to teleport. In fact they can make travel through space quantum data concerning the spin state of an electron to another electron few 10 feet away. Thanks to this some public would believe that teleportation is actually real and that this will one day be able to work even on humans. Between time travel and teleportation, teleportation things are different. Many believe that teleportation is actually impossible as it is needs that a solid material breaks the law of space and time. A solid material must disappear to then reappear in another place. This however is actually possible and in fact some scientists finally were able to teleport a solid material, which disappeared and appeared 10 feet away from the lab. The public did not believe in this but newspapers and website articles talk about it confusing the public, I personally would not believe this as it actually is impossible for something to disappear and reappear 10 feet away from you. However it can still happen if improving the machine one day this might be able to teleport people. Time travel **Science have proven that one day in the future, it will be able to travel through time. Physicists state that time travel is possible**. This however is something that humans cannot achieve easily. Few are the things necessary to travel in time, speed and kinetic energy. Some physicists believe that by using enough

speed this last will be able to break through the kinetic energy created from the speed and finally enter in the universe of the time and space continuum. Once you are in this place you can travel through different eras. This however is what many physicists believe using speed. Time travel is discussed a lot in this period and many believe that **this one day will happen**. Time travel is actually possible and this is said from one of the best minds of the world which was Albert Einstein. Einstein believes as many other physicists that time travel is possible **thanks to** the theory of **relativity**. For Einstein the speed needed to time travel must be very close to the speed of light, which is currently impossible, something must be fast near this speed, 299 792 458 m / s Since this was said to the public, this last right now believe that it will be impossible unless a machine fast enough is built. **The pace with which technology has been progressing in the last couple of years would make it undoubtable to see teleportation and time travel as the next big things**. Today's world is mainly focusing on digital transformation with topics covering cloud computing, blockchain and most importantly artificial intelligence. Time will reveal what's next on the technology roadmap.

Time travel collapses the universe.

Steve **Bowers 16**, Control Officer in the United Kingdom, Executive Editor and Moderator of the Orion's Arm Universe Project, Contributing Author for the Orion's Arm Novella Collection, ["WHY NO TIME TRAVEL IN OA", 1-1, <https://orionsarm.com/page/77>]

If the universe does allow reverse time travel, usable by sentient/sophont entities, it won't stop at one or two little historical research trips . . . If there is no effective chronological protection mechanism, **the universe of today will be overrun with travellers from the future**. Even if there is no 'Big Rip' where **the Universe tears itself apart through accelerating expansion**, hundreds of trillions of years from now the cosmos will be a slowly dying place. Even red dwarf stars will eventually burn out, leaving the inhabitants of the far future only their dying embers to gather energy from, although the creation and merger of black holes could perhaps keep civilisation going for an (admittedly very long) while. Eventually the entities of the far future will be limited to reversible computation to save energy. This means confining themselves to a very limited set of mental processes. This prospect would surely not appeal to the heirs of once-mighty advanced civilisations. If time travel were possible then **refugees from the far future would flood back**, sometimes **in multiple instances**. The future sophonts would come back in an exponentiating wave **to constantly change the present and the past**, and whole **galaxies of material particles will begin to exist in space time reference that did not have them before** - some? many? most? matter and events may turn out to be acausal, going round and round in closed timelike loops and increasing the total mass of the universe, which may begin to collapse in the distant future, sending chronistic refugees in massive tardises back to our time thus accelerating the collapse; **increasing the mass of the present day universe until it collapses**. The collapse will get closer to the present day, until it eventually happened yesterday and we will cease to exist . . . believe me, you don't want to go there. For an explanation how under certain circumstances a wormhole can connect different parts of the universe without causing temporal paradoxes see this page.

Nanotech will consume all matter by self replicating

Hu 18 – Jiaqi Hu, Humanities Scholar and President and Chief Scientist of the Beijing Jianlei International Decoration Engineering Company and 16Lao Group, Graduate of Dongbei University, Elected as the Chinese People's Consultative Conference Member for Beijing Mentougou District, Saving Humanity: Truly Understanding and Ranking Our World's Greatest Threats, p. 208-210

As a unit of measurement, a nanometer is 10^{-9} meters (or one billionth of a meter); it is roughly one 50,000th of a strand of hair and is commonly used in the measuring of atoms and molecules. In 1959, Nobel Prize winner and famous physicist Richard Feynman first proposed in a

lecture entitled "There's Plenty of Room at the Bottom" that **humans might be able to create molecule-sized micro-machines in the future and that it would be another technological revolution.** At the time, Feynman's ideas were ridiculed, but subsequent developments in science soon proved him to be a true visionary. In 1981, scientists developed the scanning tunneling microscope and finally reached nano-level cognition. In 1990, IBM scientists wrote the three letters "IBM" on a nickel substrate by moving thirty-five xenon atoms one by one, demonstrating that nanotechnology had become capable of transporting single atoms. Most of the matter around us exists in molecule forms, which are composed of atoms. The ability to move atoms signaled an ability to perform marvelous feats. For example, we could move carbon atoms to form diamonds, or pick out all the gold atoms in low-grade gold mines. However, nanotechnology would not achieve any goals of real significance if solely reliant on manpower. There are hundreds of millions of atoms in a needle-tip-sized area—even if a person committed their life to moving these atoms, no real value could be achieved. Real breakthroughs in nanotechnology could only be produced by nanobots. Scientists imagined building molecule-sized robots to move atoms and achieve goals; these were nanobots. On the basis of this hypothesis, scientists further postulated the future of nanotechnology; for example, nanobots might be able to enter the bloodstream and dispose of cholesterol deposited in the veins; nanobots could track cancer cells in the body and kill them at their weakest moment; nanobots could instantly turn newly-cut grass into bread; nanobots could transform recycled steel into a brand new car in seconds. In short, the future of nanotechnology seemed incredibly bright. This was not the extent of nanotechnology's power. Scientists also discovered that nanotechnology could change the properties of materials. In 1991, when studying C60, scientists discovered carbon nanotubes (CNTs) that were only a few nanos in diameter. The carbon nanotube became known as the king of nano materials due to its superb properties; scientists believed that it would produce great results when applied to nanobots. Later, scientists also developed a type of synthetic molecular motor that derived energy from the high-energy adenosine triphosphate (ATP) that powered intracellular chemical reactions. The success of molecular motor research solved the core component problem of nano machines; any molecular motor grafted with other components could turn into a nano machine, and nanobots could use them for motivation. In May 2004, American chemists developed the world's first nanobot: a bipedal molecular robot that looked like a compass with ten-nanometer-long legs. This nanobot was composed of DNA fragments, including thirty-six base pairs, and it could "stroll" on plates in the laboratory. In April 2005, Chinese scientists developed nano-scale robotic prototypes as well. In June of 2013, the Tohoku University used peptide protein micro-tablets to successfully create nanobots that could enter cells and move on the cell membrane. In July 2017, researchers at the University of Rome and the Roman Institute of Nanotechnology announced the development of a new synthetic molecular motor that was bacteria-driven and light-controlled. The next step would be to get nanobots to move atoms or molecules. Compared to the value produced by a nanobot, they are extremely expensive to create. The small size of nanobots means that although they can accomplish meaningful tasks, they are often very inefficient. Even if a nanobot toiled day and night, its achievements would only be calculated in terms of atoms, making its practical total attainment relatively small. Scientists came up with a solution for this problem. They decided to prepare two sets of instructions when **programming nanobots.** The first set of instructions would set out tasks for the nanobot, while the second set would **order the nanobot to self-replicate.** Since **nanobots are capable of moving atoms and are themselves composed of atoms, self-replication would be fairly easy.** One nanobot could replicate into ten, then a hundred, and then a thousand . . . **billions** could be replicated **in a short** period of **time.** This army of nanobots would greatly **increase their efficiency.** One troublesome question that arises from this scenario is: how would nanobots know when to stop self-replicating? Human bodies and all of Earth are composed of atoms; the **unceasing replication of nanobots could easily swallow humanity** and the entire planet. If these nanobots were accidentally transported to other planets by cosmic dust, the same fate would befall those planets. This is a truly terrifying prospect. Some scientists are confident that they can control the situation. They believe that it is possible to design nanobots that are programmed to self-destruct after several generations of replication, or even nanobots that only self-replicate in specific conditions. For example, a nanobot that dealt with garbage refurbishing could be programmed to only self-replicate around trash using trash. Although these ideas are worthy, they are too idealistic. Some more rational scientists have posed these questions: **What would happen if nanobots malfunctioned and did not terminate their self-replication? What would happen if scientists accidentally forgot to add self-replication controls during programming? What if immoral scientists purposefully designed nanobots that would not stop self-replicating?** Any one of the above scenarios would be enough to destroy both humanity and Earth. Chief scientist of Sun Microsystems, Bill Joy, is a leading, world-renowned scientist in the computer technology field. In April of 1999, he pointed out that if misused, nanotechnology could be more devastating than nuclear weapons. If nanobots self-replicated uncontrollably, **they could** become the cancer that **engulfs the universe.** If we are not careful, nanotechnology might **become the Pandoras box that destroys the entire universe** and all of humanity with it. We all understand that one locust is insignificant, but hundreds of millions of locusts can destroy all in their path. If self-replicating nanobots are really achieved in the future, it might signify the end of humanity. If that day came, nothing could stop unethical scientists from designing nanobots that suited their immoral purposes. Humans are *not far* from mastering nanotechnology. The extremely tempting prospects of nanotechnology have *propelled* research of nanobots and nanotechnology. The major science and technology nations have *devoted* particular efforts to this field.

Universe extinction outweighs!

Even a small chance of Universe destruction outweighs certain human extinction.

Earth is cosmically insignificant.

Hughes 18 – Dr. Nick Hughes, Postdoctoral Research Fellow at University College Dublin, PhD in

Philosophy from University of St Andrews & University of Oslo, and Dr. Guy Kahane, Professor of Philosophy at the University of Oxford, D. Phil. in Philosophy from Oxford University, "Our Cosmic Insignificance", 7-6,
<http://www.unariunwisdom.com/our-cosmic-insignificance/>

Humanity occupies a **very small** place in an **unfathomably vast** Universe. Travelling at the speed of light – 671 million miles per you are herehour – it would take us 100,000 years to cross the Milky Way. But we still wouldn't have gone very far. Our modest Milky Way galaxy contains 100–400 billion stars. This isn't very much: according to the latest calculations, the observable universe contains around 300 sextillion stars. By recent estimates, our Milky Way galaxy is just one of 2 trillion galaxies in the observable Universe, and the region of space that they occupy spans at least 90 billion light-years. If you imagine Earth shrunk down to the size of a single grain of sand, and you imagine the size of that grain of sand relative to the entirety of the Sahara Desert, you are still nowhere near to comprehending how infinitesimally small a position we occupy in space. The American astronomer Carl Sagan put the point vividly in 1994 when discussing the famous 'Pale Blue Dot' photograph taken by Voyager 1. Our planet, he said, is nothing more than 'a mote of dust suspended in a sunbeam'. Stephen Hawking delivers the news more bluntly. We are, he says, "just a chemical scum on a moderate-sized planet, orbiting round a very average star in the outer suburb of one among a hundred billion galaxies." And that's just the spatial dimension. The observable Universe has existed for around 13.8 billion years. If we shrink that span of time down to a single year, with the Big Bang occurring at midnight on 1 January, the first Homo sapiens made an appearance at 22:24 on 31 December. It's now 23:59:59, as it has been for the past 438 years, and at the rate we're going it's entirely possible that we'll be gone before midnight strikes again. The Universe, on the other hand, might well continue existing forever, for all we know. Sagan could have added, then, that our time on this mote of dust will amount to nothing more than a blip. In the grand scheme of things we are very, very small. For Sagan, the Pale Blue Dot underscores our responsibility to treat one another with kindness and compassion. But reflection on the vastness of the Universe and our physical and temporal smallness within it often takes on an altogether darker hue. If the Universe is so large, and we are so small and so fleeting, doesn't it follow that we are utterly insignificant and inconsequential? This thought can be a spur to nihilism. If we are so insignificant, if our existence is so trivial, how could anything we do or are – our successes and failures, our anxiety and sadness and joy, all our busy ambition and toil and endeavour, all that makes up the material of our lives – how could any of that possibly matter? To think of one's place in the cosmos, as the American philosopher Susan Wolf puts it in 'The Meanings of Lives' (2007), is 'to recognise the possibility of a perspective ... from which one's life is merely gratuitous'. The sense that we are somehow insignificant seems to be widely felt. The American author John Updike expressed it in 1985 when he wrote of modern science that: We shrink from what it has to tell us of our perilous and insignificant place in the cosmos ... our century's revelations of unthinkable largeness and unimaginable smallness, of abysmal stretches of geological time when we were nothing, of supernumerary galaxies ... of a kind of mad mathematical violence at the heart of the matter have scorched us deeper than we know. In a similar vein, the French philosopher Blaise Pascal wrote in *Pensées* (1669): When I consider the short duration of my life, swallowed up in an eternity before and after, the little space I fill engulfed in the infinite immensity of spaces whereof I know nothing, and which know nothing of me, I am terrified. The eternal silence of these infinite spaces frightens me. Commenting on this passage in *Between Man and Man* (1947), the Austrian-Israeli philosopher Martin Buber said that Pascal had experienced the 'uncanniness of the heavens', and thereby came to know 'man's limitation, his inadequacy, the casualness of his existence'. In the film *Monty Python's The Meaning of Life* (1983), John Cleese and Eric Idle conspire to persuade a character, played by Terry Gilliam, to give up her liver for donation. Understandably reluctant, she is eventually won over by a song that sharply details just how comically inconsequential she is in the cosmic frame. Even the relatively upbeat Sagan wasn't, in fact, immune to the pessimistic point of view. As well as viewing it as a lesson in the need for collective goodwill, he also argued that the Pale Blue Dot challenge 'our posturings, our imagined self-importance, and the delusion that we have some privileged position in the Universe'.

When we reflect on the vastness of the universe, our humdrum cosmic location, and the inevitable future demise of humanity, our lives can seem utterly insignificant. As we complacently go about our little Earthly affairs, we barely notice the black backdrop of the night sky. Even when we do, we usually see the starry skies as no more than a pleasant twinkling decoration. This sense of cosmic insignificance is not uncommon; one of Joseph Conrad's characters describes one of those dewy, clear, starry nights, oppressing our spirit, crushing our pride, by the brilliant evidence of the awful loneliness, of the hopeless obscure insignificance of our globe lost in the splendid revelation of a glittering, soulless universe. I hate such skies. The young Bertrand Russell, a close friend of Conrad, bitterly referred to the Earth as "the petty planet on which our bodies impotently crawl." Russell wrote that: Brief and powerless is Man's life; on him and all his race the slow, sure doom falls pitiless and dark. Blind to good and evil, reckless of destruction, omnipotent matter rolls on its relentless way... This is why Russell thought that, in the absence of God, we must build our lives on "a foundation of unyielding despair." When we consider ourselves as a mere dot in a vast universe, when we consider ourselves in light of everything there is, nothing human seems to matter. **Even the worst human tragedy may seem to deserve no cosmic concern.** After all, we are fighting for attention with an **incredibly vast totality.** **How could this tiny speck of dust deserve even a fraction of attention, from that universal point of view?** This is the image that is evoked when, for example, Simon Blackburn writes that "to a witness with the whole of space and time in its view, nothing on the human scale will have meaning". Such quotations could be easily multiplied—we find similar remarks, for example, in John Donne, Voltaire, Schopenhauer, Byron, Tolstoy, Chesterton, Camus, and, in recent philosophy, in Thomas Nagel, Harry Frankfurt, and Ronald Dworkin. The bigger the picture we survey, the smaller the part of any point within it, and the less attention it can get... When we try to imagine a viewpoint encompassing the entire universe, humanity and its concerns seem to get completely swallowed up in the void. **Over the centuries, many have thought it absurd to think that we are the only ones.** For example, Anaxagoras, Epicurus, Lucretius, and, later, Giordano Bruno, Huygens and Kepler were all confident that **the universe is teeming with life.** Kant was willing to bet everything he had on the existence of intelligent life on other planets. And **we now know that there is a vast multitude of Earth-like planets even in our own little galaxy.**

The best estimate is there are half a trillion current alien civilizations Lichfield 16 – Gideon Lichfield, Editor-in-Chief of MIT Technology Review, Senior Editor at Quartz, Fellow at the Data & Society Research Institute, MSc in the Philosophy of Science from the London School of Economics and Political Science, BSc in Physics and Philosophy from the University of Bristol, Former Adjunct Professor in the Global Journalism Program at New York University, "There Have Probably Been Trillions Of Alien Civilizations, And Yet We May Still Never See One", Quartz, 6-11, <https://qz.com/704687/there-have-probably-been-trillions-of-alien-civilizations-and-yet-we-may-still-never-see-one/>

[Paper internally quoted is by Adam Frank, Professor of Physics and Astronomy at the University of Rochester and Woodruff Sullivan, Professor of Astronomy and Astrobiology at the University of Washington] Sorry, everybody. **We're just not that special.** In more than five decades of scanning the heavens, the search for extraterrestrial intelligence (SETI) has found no sign of alien life. Yet now two American **astronomers,** in the scientific equivalent of a back-of-the-envelope calculation, **are estimating** that over the course of its history **the universe has seen at least half a trillion technologically advanced species.** The paper in Astrobiology by Adam Frank and Woodruff Sullivan notes that, **in just the last few years,** we've gained a much clearer sense of how hospitable the universe is to life. NASA's Kepler space telescope has identified thousands of planets in our neighborhood of the galaxy, along with their sizes and distances from their stars. From there it's fairly easy to guess how many may hold liquid water, which is probably essential for complex life. In our Milky Way galaxy alone

there are, by this estimate, some 60 billion such “habitable” planets, write Frank and Sullivan. The big remaining unknown is how many of these planets give rise to the kinds of lifeforms that build advanced technology (if nuclear weapons and Oculus Rifts can be called “advanced”). Since Earth is the only one we know of, the guesses vary wildly, but one such civilization per 10 billion habitable planets is generally considered “highly pessimistic,” wrote Frank in the New York Times yesterday (paywall). In astronomy-speak, this means the figure could be 10, 100 or even 1,000 times too low. Using that “pessimistic” proportion, and other numbers from Frank and Sullivan’s paper, I calculated how many alien civilizations should have emerged within various subregions of the universe during its history:

Alien lives should be valued as equal to humans---anything else is arbitrary and a logic of devaluation that is at the root of violence

Packer 7 – Joe Packer, then MA in Communication from Wake Forest University, now PhD in Communication from the University of Pittsburgh and Professor of Communication at Central Michigan University, Alien Life in Search of Acknowledgment, p. 62-63

Once we hold alien interests as equal to our own we can begin to reevaluate areas previously believed to hold no relevance to life beyond this planet. A diverse group of scholars including Richard Posner, Senior Lecturer in Law at the University of Chicago, Nick Bostrom, philosophy professor at Oxford University, John Leslie philosophy professor at Guelph University and Martin Rees, Britain’s Astronomer Royal, have written on the emerging technologies that threaten life beyond the planet Earth. Particle accelerators labs are colliding matter together, reaching energies that have not been seen since the Big Bang. These experiments threaten a phase transition that would create a bubble of altered space that would expand at the speed of light killing all life in its path. Nanotechnology and other machines may soon reach the ability to self replicate. A mistake in design or programming could unleash an endless quantity of machines converting all matter in the universe into copies of themselves. Despite detailing the potential of these technologies to destroy the entire universe, Posner, Bostrom, Leslie, and Ree’s only mention of alien life in their works is in reference to the threat aliens pose to humanity. The rhetorical construction of otherness only in terms of the threats it poses, but never in terms of the threat one poses to it, has been at the center of humanity’s history of genocide, colonization, and environmental destruction. Although humanity certainly has its own interests in reducing the threat of these technologies evaluating them without taking into account the danger they pose to alien life is neither appropriate nor just. It is not appropriate because framing the issue only in terms of human interests will result in priorities designed to minimize the risks and maximize the benefits to humanity, not all life. Even if humanity dealt with the threats effectively without referencing their obligation to aliens, Posner, Bostrom, Leslie, and Ree’s rhetoric would not be “just,” because it arbitrarily declares other life forms unworthy of consideration. A framework of acknowledgement would allow humanity to address the risks of these new technologies, while being cognizant of humanity’s obligations to other life within the universe. Applying the lens of acknowledgment to the issue of existential threats moves the problem from one of self destruction to

universal genocide. This may be the most dramatic example of how refusing to extend acknowledgment to potential alien life can mask humanity's obligations to life beyond this planet.

1NC

Jash its england not whales btw.

OV

A moratorium does nothing but stifle innovation.

Jennifer **Huddleston** & Christopher **Gardner 26**, *Adjunct Professor of Law at George Mason University, Senior Fellow in Technology Policy at the Cato Institute, J.D. from the University of Alabama, **Research Associate in Technology Policy at the Cato Institute, B.A. in Economics, Political Philosophy, Policy, and Law, from the University of Virginia, “Turning a Local Issue into a Federal Headache: The Problem with Data Center Moratoriums,” Cato Institute, 7/1/2026, <https://www.cato.org/blog/turning-local-issue-federal-headache-problem-data-center-moratoriums>, cameron recut jerry

A new anti-data-center push seems to be uniting the left and the right, with critics ranging from Sen. Bernie Sanders and Rep. Frank Pallone to Sen. Josh Hawley and Rep. Nancy Mace calling for some form of federal limitation on data center construction. As with any expansion, there are legitimate policy conversations to have about resource use and other underlying issues. However, a moratorium on the construction of a particular kind of facility based on arbitrary ceilings when demand is growing is a misguided policy sledgehammer that could not only derail critical innovation but also fail to resolve the underlying policy issues.

Data centers aren’t new, but the rising backlash and policy proposals to ban construction come at a critical time for innovation. The need for data centers should not be seen as an either/or but rather as an opportunity to consider better solutions to policy questions around resource use and other related issues facing the communities where they are built. If we aren’t careful, bad policies like data center construction moratoriums and outdated energy policy could backfire – not only on progress in artificial intelligence, but also on a range of current and future technologies, including cloud computing and quantum computing.

The Rise of Data Center Construction Moratorium Policy and Its Problems

The most concrete federal effort to block data center construction is the AI Data Center Moratorium Act. Proposed by Senator Bernie Sanders on March 25, 2026, this Act would halt construction of new or existing data centers until a comprehensive federal AI regulatory framework is enacted. In the months since, the Overton window has shifted, with both the public and policymakers seemingly expressing a desire to pause construction of critical infrastructure for current and emerging technology demands.

The federal government is not alone. As of the end of June 2026, 116 municipalities have imposed local moratoriums. At the state level, California, Minnesota, Montana, and Kansas have all passed legislation that either incentivizes or disincentivizes data center construction in their states. These are in addition to the countless AI bills proposed in state legislatures across the country.

Unsurprisingly, states and localities have been and will continue to both restrict and enable data center construction for the foreseeable future.

Responding to Critics’ Common Concerns

Senator Sanders justified his bill by arguing two main points common among data center critics. His first claim is that AI is advancing too quickly and that Congress is moving too slowly. Because of this, Congress needs a data center moratorium to give it time to catch up with the technology. His second claim is that data centers pose environmental and economic risks to their locality by raising utility rates and consumption.

The data on data centers, however, do not bear out these claims. Data centers are needed to meet increasing demands beyond just AI, including the volume of connected devices and data Americans use regularly. Our capacity needs have expanded with AI, but demand had been growing even before AI's exponential leaps. Data centers are critical to AI and the progress it enables across a variety of areas. AI products power not only the direct-to-consumer chatbots many have come to rely on to improve access to information but also an array of products that can improve health, enhance cybersecurity, or support responses to natural disasters. All of this is to say AI's benefits are far from merely ephemeral.

When it comes to concerns about energy usage or utility rates, there are better solutions than the moratorium sledgehammer, and the data indicate such claims are vastly overblown. Multiple studies have found that while energy rates have increased, it's far from clear that data centers are to blame. In fact, one study even found a connection between lower retail energy rates and data centers.

The reality is that energy policy keeps the grid and our energy supply outdated, which can increase costs. A better solution is to pursue energy policy reform, as our colleague Travis Fisher has written, to enable greater energy abundance and innovation.

Similarly, water usage in some regions of the country is a legitimate debate. But it is a problem that is far broader than just data centers. In aggregate, golf courses use significantly more water than data centers but rarely face the same kind of backlash.

Why The Backlash Could Backfire

The reality is that technological advancement is overwhelmingly a force for good. Some communities don't want a new data center, while others may embrace the millions in tax revenue and hundreds of high-paying jobs that come with it. Individual communities are best positioned to know the specific trade-offs they face with any large investment, as they have been for decades.

Data centers date back about 70 years to when IBM was building them to support the cutting-edge technology of the day: mainframe computers. The type of data centers we think of today can trace back over 25 years. Cities and regions like Loudon County, Virginia, have worked with communities and companies to find a balance that enables infrastructure investments that provide economic opportunities and development while responding to community-specific concerns. More recently, however, these debates have spiraled from the typical debates over new construction or development to something much more heated.

The backlash against datacenters and AI has, at times, mushroomed into irrationality and violence. On April 10, 2026, a Molotov cocktail was thrown at Sam Altman's home. Two days later, the same home was struck by gunfire from a vehicle. In 2025, a 25-year-old was arrested and charged with an act of terrorism for his plan to burn down xAI's Memphis Colossus datacenter. These are paired with shootings at the homes of local councilmembers, physical confrontations at town meetings, and death threats against county commissioners.

This is particularly concerning, given the often-overlooked benefits of data centers. One study estimated that data center construction could create 4.7 million construction and skilled trade jobs in the United States. Many skilled trade jobs last beyond the initial construction, given the increased need for HVAC technicians and electricians, among other jobs, to support the center's operation.

For users, there is a growing demand for less latency and more storage. The downside is that without the necessary infrastructure to support the demands of modern startups, it grows more difficult to start the next great company in a garage. This could also mean that those needing this level of access to develop cutting-edge AI applications must migrate to regions that provide the necessary data center infrastructure to support it.

This is in part why a federal moratorium would be particularly concerning. Companies and talent need to know they will not face unnecessary roadblocks from the government, particularly for capital-intensive investments. Given the global nature of AI competition, preventing such a critical buildout could hinder the US's global competitiveness.

1st Off

Interpretation: Debaters must disclose all previously read non-identity constructive positions open source with highlighting, tags, and any paraphrased text on the 2026-2027 NDCA PF wiki under their institution and team at least 30 mins before the round or right after pairings come out

Violation: They don't disclose--inserted screenshots prove their page

West Windsor Plainsboro South HaTe (Harish-Telang), 9/11/2026, OpenCaseList,
<https://opencaselist.com/hspf26/WestWindsorPlainsboroSouth/HaTe>

The screenshot shows a web browser window with the OpenCaseList website. The page title is "West Windsor Plainsboro South HaTe (Harish-Telang)". The page is for the 2026-2027 season, specifically for the HS PF 2026-27 round. The page is currently empty, showing "No rounds yet, add one!" under the "Round" column. The "Team Notes" section is also empty. The "Team History" section is empty. The left sidebar shows a list of schools, including Acton-Boxborough (MA), ADL (TX), American (CA), American Heritage Broward (FL), American Heritage Palm Beach (FL), Ansh And Friends (TX), Apollo (GA), Archbishop Mitty (CA), Ardrey Kell (NC), Arizona Chandler Independent (AZ), Arizona College Prep (AZ), Arlington (MA), Ascent Academy (TX), Auburn (AL), Austin Peace Academy (TX), Awty (TX), Awty Honors Independent (TX), Bartlesville (OK), BASIS Independent Fremont (CA), BASIS Peoria (AZ), BASIS Phoenix (AZ), BASIS San Antonio Shavano (TX), BASIS Scottsdale (AZ), Belen Jesuit Prep (FL), Bellaire (TX), Bellarmine (CA), Bellarmine College Prep (CA), Bellevue (WA), Bergen County Academics (NJ), Bergen Debate Club (NJ), Bergen Tech (NJ), and Blake (MN). The bottom of the page shows a Windows taskbar with various icons and the system clock indicating 8:03 PM on 9/11/2026.

Case

The role of the ballot is determine the truth or falsity of the resolution

[1] Constitutive: The ballot asks you to either vote aff or neg based on the given resolution--its intrinsic to the activity and to stop intervention

Affirm means “assert to be true,” 5 dictionaries.

Dictionary.com n.d. [<https://www.dictionary.com/browse/affirm>]

affirm [uh-furm] **verb** (used with object) to **state or assert** positively; **maintain as true**: to affirm one's loyalty to one's country; He affirmed that all was well. Synonyms: testify, depose, asseverate, aver
Antonyms: deny

Negate means “deny the truth of.” 5 dictionaries:

Dictionary.com n.d. [<https://www.dictionary.com/browse/negate>]

verb (used with object) , ne-gat-ed, ne-gat-ing. **to deny the** existence, evidence, or **truth of**: an investigation tending to negate any supernatural influences. to nullify or cause to be ineffective: Progress on the study has been negated by the lack of funds.

[3] Regress: Every ROB requires saying something is true or false in order to provide a win condition, but the resolution is the only non-arbitrary stasis point.

‘Should’ is mandatory AND express moral duty.

Keller ’32 [W.H.; July 14; Judge on the Superior Court of Pennsylvania; Westlaw, “Foresi v. Hudson Coal Co.,” 106 Pa.Super. 307, <https://www.casemine.com/judgement/us/5914a5c2add7b049346cf8d5>]

As regards **the mandatory character** of the rule, the **word “should”** is **not only an auxiliary verb**, it is also **the preterite of the verb, “shall,”** and has for one of its meanings, as defined in the Century Dictionary: “**Obliged or compelled (to); would have (to); must; ought (to); used with an infinitive (without to) to express obligation, necessity or duty** in connection with some act **yet to be carried out**.” We think it **clear** that **it is in that sense** that **the word “should”** is used in this rule, **not merely advisory**. When the judge in charging a jury tells them that, unless they find from all the evidence, beyond a reasonable doubt, that the defendant is guilty of the offense charged, they should acquit, **the word “should”** is **not used in an advisory sense**, but has **the force or meaning of “must,”** or **“ought to,”** and carries with it the sense of *313 **obligation and duty equivalent to compulsion**. A natural sense of sympathy for a few unfortunate claimants who have been injured while doing something in direct violation of law must not be so indulged as to fritter away, or nullify, provisions which have been enacted to safeguard and protect the welfare of thousands who are engaged in the hazardous occupation of mining.

There is no free will. Human action follows a deterministic path- most recent science

proves.

Coyne 12 [Jerry Coyne, Professor in the Department of Ecology and Evolution at The University of Chicago, "Why You Don't Really Have Free Will," USA Today, January 1st, 2012

<https://www.ethicalpsychology.com/2013/12/why-you-dont-really-have-free-will.html?m=1>]

The first is simple: we are biological creatures, collections of molecules that must obey the laws of physics. All the success of science rests on the regularity of those laws, which determine the behavior of every molecule in the universe. Those molecules, of course, also make up your brain — the organ that does the "choosing." And the neurons and molecules in your brain are the product of both your genes and your environment, an environment including the other people we deal with. Memories, for example, are nothing more than structural and chemical changes in your brain cells. Everything that you think, say, or do, must come down to molecules and physics. True "free will," then, would require us to somehow step outside of our brain's structure and modify how it works. Science hasn't shown any way we can do this because "we" are simply constructs of our brain. We can't impose a nebulous "will" on the inputs to our brain that can affect its output of decisions and actions, any more than a programmed computer can somehow reach inside itself and change its program.

Criterion

Criterion: There's no way to determine what's good or bad, because that requires a method for determining itself.

Roderick **Chisholm 82**. Professor of Philosophy at Brown University from 1947 to 199. 1982. University of Minnesota Press. "The Foundations of Knowing." Chapter 5, The Problem of the Criterion. https://drive.google.com/file/d/1EVZ82GKHmf29AA_bAI5_LNc0ttOPOAjV/view?usp=sharing.

The question we started with was: How are we to tell the good ones [beliefs] from the bad ones [beliefs]? In other words, we were asking: What is the proper method for deciding which are the good beliefs and which are the bad ones, which beliefs are genuine cases of knowledge and which beliefs are not?

And now, you see, we are on the wheel. First, we want to find out which are the good beliefs and which are the bad ones. To find this out we have to have some way, some method of deciding which are the good ones and which are the bad ones. But there are good and bad methods good and bad ways of sorting out the good beliefs from the bad ones. And so we have a new problem: How are we to decide which are the good methods and which are the bad ones?

If we could fix on a good method for distinguishing between good and bad methods, we might be all set. But this, of course, just moves the problem to a different level. How are we to distinguish between a good method [and a bad method] for choosing good methods? If we continue in this way, of course, we are led to an infinite regress and we will never have the answer to our original question.

Pain/Pleasure Bad

Darwinian Dilemma: Pain and pleasure aren't a binding source for ethics.

Sharon **Street 06**. Professor of Philosophy and Associate Chair of the Department of Philosophy at New York University. "A Darwinian Dilemma for Realist Theories of Value." *Philosophical Studies: An International Journal for Philosophy in the Analytic Tradition*. Vol. 127, No. 1, pp. 109-166, <https://www.jstor.org/stable/pdf/4321684.pdf>.

Suppose, however, there is remaining doubt and some are still tempted by a realist position on the badness of pain so understood. It's at this point that the Darwinian Dilemma arises again for the realist. To see how, suppose again that pain is given some definition according to which it is not a necessary feature of pain that we unreflectively experience it as counting in favor of what would avoid, lessen, or stop it. In that case, the following becomes a legitimate scientific question: given that it is perfectly conceivable that we all could have ended up taking pain sensations to count in favor of what would cause them and intensify them rather than in favor of what would lessen them and stop them, what explains the fact that such a huge percentage of us so consistently do the latter? Here, as in earlier cases, there is a powerful evolutionary answer. I've left it open how the person opting for the first horn of the Pain Dilemma is defining pain (so long as that definition makes no reference to the idea that pain is a sensation that we unreflectively take to count in favor of what would stop it). But if the proposed definition is to be plausible at all, then it will pick out (predominantly, one assumes) sensations associated with the sorts of bodily conditions that we normally consider painful, such as cuts, burns, bruises, broken bones, and so on. And it is of course no mystery whatsoever, from an evolutionary point of view, why we and the other animals came to take the sensations associated with bodily conditions such as these to count in favor of what would avoid, lessen, or stop them rather than in favor of what would bring about and intensify them. One need only imagine the reproductive prospects of a creature who relished and sought after the sensations of its bones breaking and its tissues tearing; just think how many descendants such a creature would leave in comparison to those who happened to abhor and avoid such sensations.

As in earlier cases, the realist faces a problem when confronted with such an explanation. For once again we see that there is a striking coincidence between the content of the independent evaluative truth posited by the realist, on the one hand, and the content that evolutionary theory would lead us to expect, on the other. The realist tells us that it is an independent evaluative truth that pain sensations (however he or she defines them) are bad, and yet this is precisely what evolutionary theory would have predicted that we come to think. And once again the realist is unable to give any good account of this coincidence. To insist that the coincidence is mere coincidence is implausible. The realist's alternative, here as in earlier cases, is to defend some sort of tracking account, according to which we were selected to be able to discern independent evaluative truths, among them the truth that these pain sensations (however the realist is defining them) are bad. Yet here as in earlier cases, the tracking account is scientifically unacceptable. In order to explain why we came to think that these sensations are bad, we need make no reference whatsoever to the fact that they are bad; we need only point out how it tended to promote reproductive success to take them to be bad (due to their connection with bodily conditions that tended to diminish reproductive success).

The realist, then, is forced to the other horn of the Pain Dilemma. To salvage realism about the badness of pain, he or she is forced to understand pain as a sensation such that the creature who has it unreflectively takes that sensation to count in favor of whatever would avoid, lessen, or stop it. But now notice what this means. In order to salvage his or her view of pain as bad independently of our evaluative attitudes, the realist must admit that pain's badness depends on its being a sensation such that the creature who has it is unreflectively inclined to take it to be bad. But this, in turn, is just to admit that its badness depends in an important sense on our evaluative attitudes in particular, on our being unreflectively inclined to take it to be bad. Pain may well be bad, in other words, but if it is so, its badness hinges crucially on our unreflective evaluative attitudes toward the sensation which pain is. The realist is thus forced to recognize the role of our evaluative attitudes in determining the disvalue of pain. Though initially plausible, it is a mistake to say that pain is bad independently of our evaluative attitudes. Pain, if it is plausibly to be construed as bad independently of our other evaluative attitudes, must be understood as a sensation such that we have a certain evaluative attitude toward it and it's that evaluative attitude which (at least in part) makes the sensation bad.

Skep Negates

1] Absolute Truth: It's necessary.

[A] Logic: It's grounded in literature.

Walter **Sinnott-Armstrong 24**. American philosopher specializing in ethics, epistemology, neuroethics, the philosophy of law, and the philosophy of cognitive science. The Chauncey Stillman Professor of Practical Ethics in the Department of Philosophy and the Kenan Institute for Ethics at Duke University. Aug 1, 2024. Stanford Encyclopedia of Philosophy Archive. "Moral Skepticism." <https://plato.stanford.edu/archives/fall2024/entries/skepticism-moral/>. Prateek

The final kind of argument derives from René Descartes (1641). I do not seem justified in believing that what I see is a lake if I cannot rule out the possibility that it is a bay or a bayou. Generalizing, **if there is any contrary hypothesis that I cannot rule out, then I am not justified in believing** that what I see is a lake. **This common standard is** supposed to be **required for justified belief**. When this principle is applied thoroughly, it leads to skepticism. **All a skeptic needs to show is that**, for each belief, **there is some contrary hypothesis that the believer cannot rule out**. It need not be the same hypothesis for every belief, but skeptics usually buy wholesale instead of retail, so they seek a single hypothesis that is contrary to all (or many common) beliefs and which cannot be ruled out in any way.

[B] Truth is an absolute burden---any risk they're wrong is a neg ballot.

Peter **Unger 78**. Contemporary American philosopher and professor in the Department of Philosophy at New York University. January 1978. Ignorance: A Case for Scepticism. "An Argument for Universal Ignorance." drive.google.com/file/d/1Fsi58NshZrSkrYrUbggLXz6Efy7bJFn4/view?usp=sharing.

The very particular idea that **knowing entails its being all right to be certain** is suggested, further, by the fact that knowing entails, at least, that one is certain. As we saw in section g of the preceding chapter, that this is a **fact is made quite plain by the inconsistency expressed by sentences like 'He really knew that it was raining, but he wasn't absolutely certain that it was.'** **Such a sentence can express no truth:** if he wasn't certain, then he didn't know. we get further confirmation from considering related sentences. To anticipate again our work in Chapter IV, the sentences 'He was sad that it was raining, but he didn't know it was' and 'He was really sad that it was raining, but he wasn't absolutely certain it was' are likewise inconsistent. Their inconsistency means an entailment from being sad that p to knowing that p, in the first case, and to being certain that p in the second. The same experience is felt with a statement to the effect that someone regretted that he did something, and with the denials that he knew he did, and that he was certain of it. This can be best explained, it would seem, by an entailment from knowing to being absolutely certain. The entailment from knowing to being certain is pretty convincingly clinched, I think, by appreciating the equivalence between someone's knowing something and his knowing it for certain, or with absolute certainty. To be sure, we may describe cases which we would more naturally react to with the words 'He knew it' than 'He knew it for certain.' One such case is that of the examined student (considered in Chapter II, section g). Another concerns a man who, looking for his cuff-links, unerringly goes to the very spot where they are while doubts go through his mind. Does he know that they are in that spot? We readily say 'Yes'. But our readiness to say that these people know might only indicate loose usage of 'know' by us, while we are more strict in our use when the words 'certain' and 'certainty' enter the picture. That this is much the more plausible hypothesis than thinking there to be an inequivalence here is evidenced by the inconsistency of the relevant sentences: 'He knew it, but he didn't know it for certain,' 'He really knew it, but he didn't know it with absolute certainty,' 'He knew it was there, but he didn't really know it', and so on. No truth can be found in these words no matter when they might be uttered. Even if they are put forth at the end of stories like that of the cuff-link finder, where we are inclined at first to say he knows, we realize that they must express what is false. Accordingly, we are forced to be unswayed by our tendency to loose usage and to admit the equivalence between knowing with absolute certainty and just plain knowing to be so. Admitting this equivalence, we can be quite confident that knowing does indeed entail being absolutely certain. This reinforces further the arguments and conclusions of the chapter just preceding.

2] Burdens: Disproving the obligatory nature of the resolution is a negative win condition.

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Burden of Proof: the affirmative's responsibility to prove that the resolution is true. If the affirmative fails to prove the resolution, they ought to lose the debate.

3] Presumption: Presumption isn't triggered since our statement logically negates the resolution making skep offense. But it negates---rulebook outweighs.

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Presumption: an argument that says that the current system works or is sufficient until proven otherwise, so the plan is not needed. Common in Policy Debate.